

# CS 2614: Computer Organization

## Lab 6

Spring 2025

### Seven-Segments Display

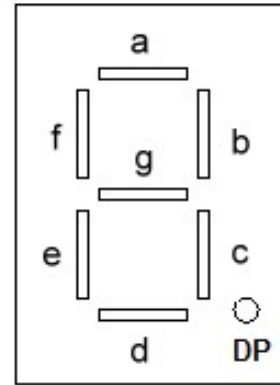
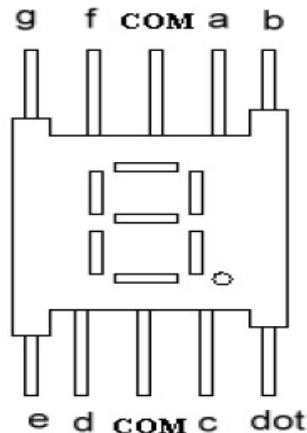
| Student ID  | Name         |
|-------------|--------------|
| 113 568 816 | Colby Frison |
| 113 570 421 | Bricen Ford  |



***The UNIVERSITY of OKLAHOMA***  
*Gallogly College of Engineering*  
School of Computer Science

## Pre-Lab Exercise

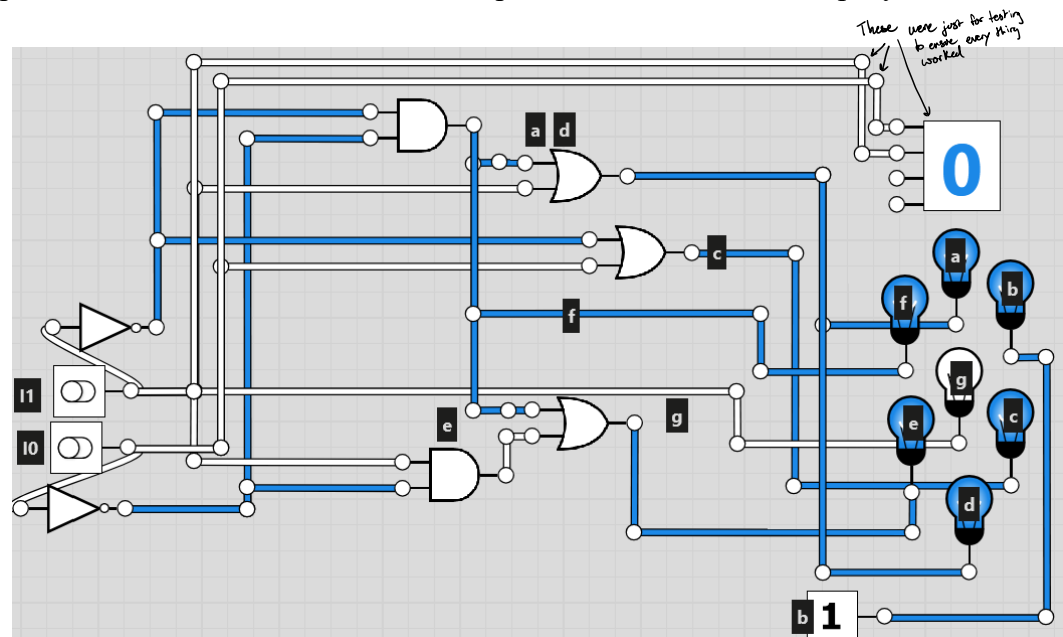
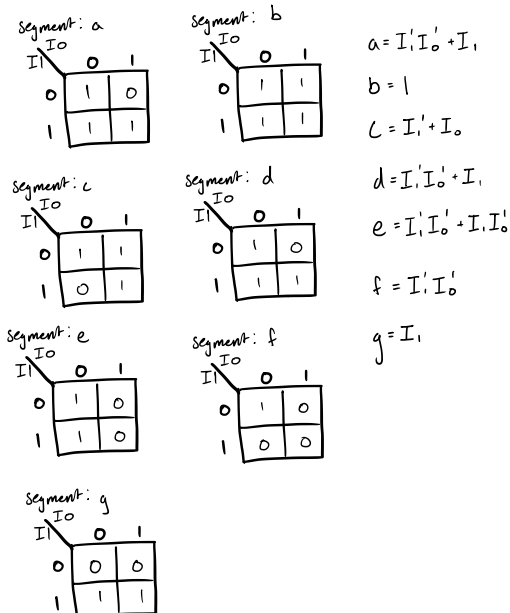
- The seven-segments display we are going to use in the lab consists of 8 LEDs (a, b, c, d, e, f, g and DP). In this exercise, we are going to design a combinational circuit such that the seven-segments display shows decimal symbols depending on the input.



- Derive the truth table for displaying decimal digits 0, 1, 2 and 3 on the seven-segment display. **Digit 1** should be on the **right side** of the seven-segment display. [Hint: For a light to be ON, it should correspond to an output of 1]

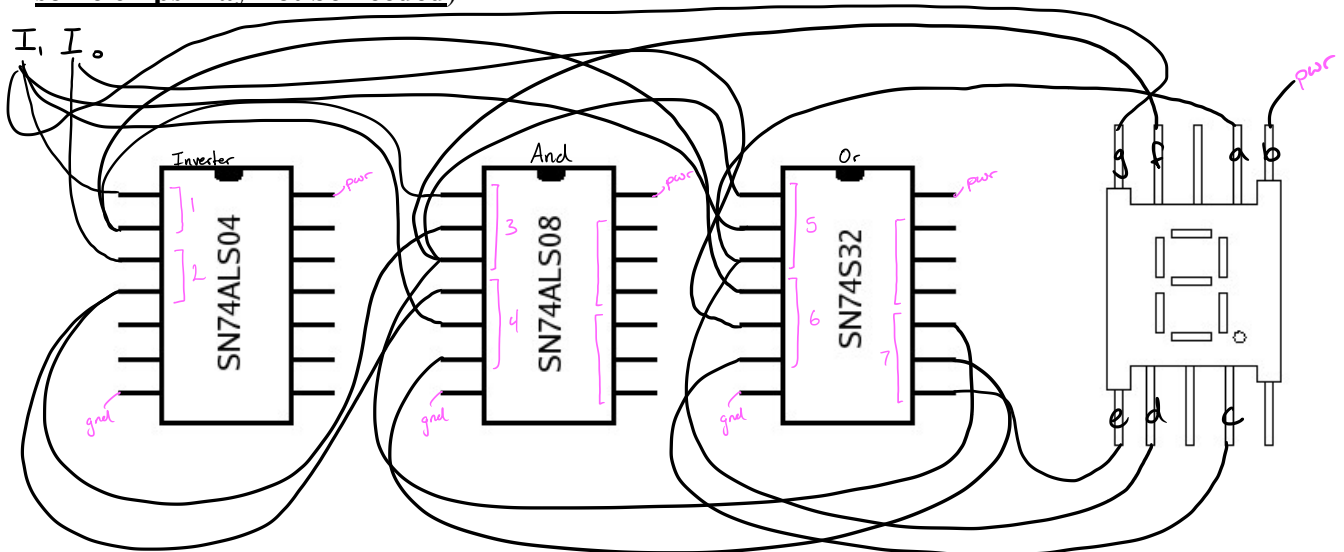
| Decimal | Inputs         |                | Outputs |   |   |   |   |   |   |
|---------|----------------|----------------|---------|---|---|---|---|---|---|
|         | I <sub>1</sub> | I <sub>0</sub> | a       | b | c | d | e | f | g |
| 0       | 0              | 0              | 1       | 1 | 1 | 1 | 1 | 1 | 0 |
| 0       | 1              | 0              | 0       | 1 | 1 | 0 | 0 | 0 | 0 |
| 1       | 0              | 1              | 1       | 1 | 0 | 1 | 1 | 0 | 1 |
| 1       | 1              | 1              | 1       | 1 | 1 | 1 | 0 | 0 | 1 |

- Convert the above truth table to a combinational circuit. [Hint: Solve for output bits (a through g) one by one by using K-map method. This will create seven K-Maps. Then, combine and simplify the results to get the circuit]



## In-Lab Exercise

1. Draw the circuit from Problem 1(b) using the chips below (label the chips and each input/output, some chips may not be needed).



2. Implement the circuit from Problem 2 on the breadboard.

